



How Might Voice Assistants Raise Our Children?

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Abstract. When mobile devices such as tablets and smartphones were becoming more popular, an important question that psychologists and pediatricians asked was how interactions facilitated by these devices with screens may affect the functioning of children. Nowadays, when technology used by children, such as intelligent voice assistants, does not require a screen at all, these issues seem to fall into the background. Today, concerns are growing about the effect of interacting with voice-driven AI services, as it may potentially have a greater impact on children's cognitive development than engaging with television or smartphones. The purpose of this paper is to outline potentially interesting directions of research in the field of voice assistant technology concerning how this solution may affect the functioning of children, and in particular if, and to what extent, it may redefine the dynamics of social contacts within and outside the family.

Keywords: Intelligent voice assistants · Human factors · Digital parenting

1 Introduction

For several years, artificial intelligence has been the subject of debate far beyond the field of technology itself, as awareness of a wide range of its social consequences has been increasing. AI technology has progressed so much that it is now possible to use it in many processes that required human intervention until recently (e.g. voice recognition or translation). AI-enhanced software is used, for example, in commerce where intelligent systems can talk to customers in call centers without human intervention and external control. Speech recognition systems are becoming widespread, along with voice assistant systems, such as Google Assistant, Siri, Alexa and Cortana. According to recent research [1], 41% of children and young people (aged 9–17) in the UK have a voice-activated personal assistant on their phone, while 23% have such a device at

home. Another 13% of children declare they “would like to have one to talk to”. As a consequence, concerns are growing about the potential impact of artificial intelligence and voice assistants on children, as it is thought to be greater than that of television or smartphones, and some groups, such as the Campaign for a Commercial Free Childhood [2], are lobbying against them. These concerns about children’s welfare arise with each new mass medium or technology being introduced to a broader public: from a 19th-century novel, through radio and TV, to computers, the Internet and, lately, mobile devices [3, 4].

In light of the above, it is vital to explore the current state of the art, highlighting both the benefits and challenges, and to present opportunities for further research relating to the use of intelligent voice assistants (IVAs) by children. This is even more relevant since the transition from interaction with screens (PCs, smartphones, tablets) to interactions with voice assistants does not merely entail a change in the mode of communication (touch/vision-based versus voice-based). This is in part because voice communication is the most natural way of communicating with other people. What follows in such human-computer interactions (HCI) is a transformation of the perceived status of the machine, which acquires human characteristics [5, 6]. This altered status may have far-reaching consequences for the cognitive aspects of such interaction, and it may necessitate a paradigm shift in HCI studies towards a more interdisciplinary approach to related research.

The purpose of this article is to pose a few open questions and to outline potentially [5, 6] interesting areas of research in the field of voice assistant technology. In particular, it asks questions on how this technology will affect children’s lives, e.g., how it may redefine the dynamics of social contacts within and outside the family. Many young children, who have not yet learnt to read and write, may have their psychological world shaped, *inter alia*, by interactions with voice assistants. These are important issues, especially considering that children are among the largest groups of users of such systems.

2 Considerations and Limitations

As with any other media/technology effects, it is important to acknowledge—and search for—individual and societal factors potentially moderating the impact of the use of IVA on children. Below we present an overview of some possible factors influencing the nature of child-IVA interaction and its potential impact on children’s cognition, based on a literature review of related topics [7, 8] and our research and experience in this area. Child-specific factors include: age and gender of the child, neurotype of the brain (if a child is neurotypical or autistic), social context of the use (e.g. in a family setting, supervised or on their own parental/social mediation), the mode of use (mostly as a Smart Home solution, or an Assistant). IVA-specific factors comprise: functionality of the IVA, gender of the IVA’s voice, level of possible interactivity, target group of the solution (developed for adults or specifically for children) and the physical casing of the device (a home speaker, a phone or a toy).

3 Promising Areas of Research

3.1 The Effect of Voice Assistants on Linguistic Habits

The activation commands for Google and Apple Home Pod voice assistants are “Hey, Google” or “Hey, Siri”, whereas interactions with Amazon’s Alexa begin simply with “Alexa...” (also “Echo” or even “Computer”). These forms, stripped of strategies introducing politeness, were chosen for brevity, but it makes them sound like commands. For instance, expressing gratitude with a ‘Thank you’ was not required until recently and the option to encourage polite exchanges was only added in the 9 May 2018 update of Alexa. Recent changes in this area came as a way to address some parents’ concerns connected to the VA impact on children’s politeness in interpersonal dealings in later life. Mattel also planned to introduce a voice assistant for children that will require them to speak politely, but they cancelled their plans due to concerns raised by child advocacy groups, lawmakers and parents [9–11]. Related research has already shown that such a requirement changes adults’ manner of speech to a more polite one [12]. This impact may be even greater because some children do not consider voice assistants to be devices but, instead, anthropomorphize them to some extent by attributing characteristics such as ‘friendly’ or ‘reliable’ [13]. It is, however, worth asking whether voice assistants might reinforce behaviors that parents would like to avoid in their children, i.e. ones that may make children’s lives more difficult in adulthood. The effectiveness of voice assistants provides positive reinforcement, reassuring children that the only condition for getting what they want is proper enunciation, whereas politeness and good manners are on the sidelines. The positive reactions of peers observing how they interact with IVAs may also strengthen this effect. All of this might lead to a situation where the child transfers such behaviors to different social contexts, such a classroom or the family environment.

3.2 Voice Assistants’ Effects on Social Relationships

During the Google I/O 2018 conference, Google presented the functionalities of its voice assistant which allow it to make phone calls in order to perform tasks assigned by the user. One of the presented examples was booking an appointment with a hairdresser. During the conversation, the person answering the call at the hairdresser’s salon was not even aware that she was speaking to an artificial being. Such presentations, which demonstrate the state-of-the-art in voice assistant technology, make us wonder to what extent intelligent voice assistants might supplement or even substitute interpersonal contacts further down the line. A few years ago, when smartphones were becoming more widespread among children and young adults, similar questions were asked as there were fears that contact with screens might negatively impact children’s cognitive, emotional, social and even physical development, e.g. suppress contact with peers, lead to physical inactivity, obesity and inadequate sleep quality [14]. Nowadays, we can see that phones or tablets with instant messaging services have merely become just another communication channel.

However, the issue of voice assistants being anthropomorphized emerges here again because communication occurs through voice, a channel associated with

interpersonal contacts. In this context, a question arises whether it is possible for voice assistants to become a new sophisticated kind of ‘imaginary friend’. It seems that this technology is exceptionally well suited for this role [15], as children themselves prefer the voice interfaces to be personified [16]. Importantly, voice assistants are deprived of all the negative features of a human interlocutor. They will patiently listen to everything, without ridiculing or revealing the secrets ‘entrusted’ to them (depending on the privacy policy and settings). The fact that a voice assistant may become a kind of a friend or a trusted person for children raises further questions, also explored in the controversial book “To Siri with Love: A Mother, Her Autistic Son, and the Kindness of Machines” [17]. Although the patience of an IVA is a major benefit for some children, the inflexible interaction may sometimes be a source of frustration for others. Overall, when communication with a VA breaks down, children employ repair strategies borrowed from their interpersonal experience, proving that they also have the patience to stay engaged despite difficulties. Here, it is important to note that the transfer of attitudes happens both ways.

Research on the use of voice interfaces in a family setting indicates that assistants can supplement family interactions, as is often the case when novel yet but accessible technology is used in intergenerational contexts of edutainment. This is true even to a point where IVA queries become “embedded in the life of a home” as Porcheron et al. [18] describe. Thus, this use may encourage socializing, with ready access to games, quizzes or other forms of entertainment.

3.3 Voice Assistants’ Effect on Children’s Everyday Habits

The results of the famous ‘marshmallow test’ and its subsequent replications and variations [19] indicate that the ability to postpone gratification is correlated with effective functioning in adolescence and adulthood. Children who were able to resist consuming the sweets immediately during those experiments demonstrated better cognitive abilities in adolescence. This can be interpreted as a signal to parents that they should strengthen their children’s ability to postpone gratification. This raises the question related to interactions with voice assistants, which produce the opposite effect and, as a result, could teach children to expect immediate responses to their requests [10], for instance, if children want to listen to their favorite songs. Obviously, we should remember that voice-led interactions are still quite complicated even for adults, and sometimes almost impossible for toddlers.

On the other hand, voice assistants may also help children stay on track with digital and shared to-do lists and chores, and it is up to the parents to learn to use these features to their advantage. Moreover, during the interaction with IVA, children are learning how to ask more specific questions, which may teach them that patience and persistence is eventually rewarded. Also, toddlers may be encouraged to improve their pronunciation and learn to speak more slowly [13].

3.4 Ethical and Moral Aspects of Voice Assistants

The questions of decision-making, control and liability always surface in the context of AI-powered technology. Apart from the underlying programming, the actual user

settings concerning voice priority, gender, the manner of speaking and censorship, there are also ethical dilemmas connected with the range of interactions, and whether it should include advice or even reporting on negative behaviors. With IVAs, the topic of privacy is vital as all voice interactions are recorded and analyzed. It is unclear whether and how a voice assistant should react when the child provides information indicating a violation of the law, e.g. on taking drugs, being harassed at school or being sexually abused. Should the system use a neural network to process and analyze the content of such utterances or should such quotes be anonymized and flagged for review by ML-supported trained professionals [20] or volunteers [21]? Perhaps neither of these things ought to happen.

Issues related to AI ethics are also widely discussed in the context of autonomous cars, where a controversy arose around the first victims of car accidents. Currently, moral discussions in this area are limited primarily to dilemmas regarding accidents that cannot be avoided (the so-called trolley problem [22]). This shows that insufficient emphasis is placed on broader questions that should have been asked at a very early design stage, e.g., “what design decisions have led to this moral dilemma,” “what kind of values should guide the project and how they should be weighed.” Once these considerations become more transparent in the specification of available commercial solutions, users would have the possibility to make informed decisions based on their moral compass and privacy preferences.

3.5 Conclusions

As the use of IVAs is becoming more common in many areas of life, it is important to consider the long-term effects it may have on social relationships including aspects such as encouraged attitudes and the use of language. These considerations are especially important when it comes to children who will be growing up surrounded by AI-enhanced technology. The possible cognitive effects of IVAs on children, especially the youngest ones, are hard to predict clearly, but the extent of their positive or negative impact will depend on many child- and solution- specific factors such as the mode of IVA use, and the moderation of caretakers. This is why it is essential to conduct further research on ways to improve IVAs, especially to support children’s development or assist them in learning and on potential functions that may help enrich family lives. Overall, when used responsibly, this technology has great potential since it enables natural and hands-free communication which can be seamlessly introduced into many environments, creating user-friendly and accessible experiences both for the older and the younger generation, encouraging them to engage with it together.

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